

integral here, as appropriate for a diagram with one loop. We shall see how to do this sort of momentum-space integral in Chapter 11.

To make this more specific, consider a theory with a Dirac spinor field $\psi(x)$ of mass M and a pseudoscalar field $\phi(x)$ of mass m , interacting through the interaction $-ig\bar{\psi}\gamma_5\psi$. (The factor $-i$ is inserted to make this interaction Hermitian for real coupling constants g .) Recall that the polynomial $P(q)$ for the scalar is just unity, while for the spinor it is $[-i\gamma_\mu q^\mu + M]\beta$. Also, the u for a scalar of energy E is $(2E)^{-1/2}$, while for the spinor u is the conventionally normalized Dirac spinor discussed in Section 5.5. Eqs. (6.3.8), (6.3.9), and (6.3.10) give the lowest-order connected S -matrix elements for fermion-boson scattering, fermion-fermion scattering, and boson-boson scattering:

$$\begin{aligned}
 S_{\mathbf{p}'_1 \sigma'_1 \mathbf{p}'_2 \sigma'_2, \mathbf{p}_1 \sigma_1 \mathbf{p}_2 \sigma_2} &= -i(2\pi)^{-2} g^2 (4E'_2 E_2)^{-1/2} \delta^4(p_1 + p_2 - p'_1 - p'_2) \\
 &\times \left[\left(\bar{u}(\mathbf{p}'_1 \sigma'_1) \gamma_5 \frac{-i\gamma_\mu (p_{1'} - p_1)^\mu + M}{(p_{1'} - p_1)^2 + m^2 - i\epsilon} \gamma_5 u(\mathbf{p}_1 \sigma_1) \right) \right. \\
 &\quad \left. + \left(\bar{u}(\mathbf{p}'_2 \sigma'_2) \gamma_5 \frac{-i\gamma_\mu (p_{2'} - p_2)^\mu}{(p_{2'} - p_2)^2 + m^2 - i\epsilon} \gamma_5 u(\mathbf{p}_2 \sigma_2) \right) \right], \\
 S_{\mathbf{p}'_1 \sigma'_1 \mathbf{p}'_2 \sigma'_2, \mathbf{p}_1 \sigma_1 \mathbf{p}_2 \sigma_2} &= i(2\pi)^{-2} g^2 \delta^4(p_1 + p_2 - p'_1 - p'_2) \\
 &\times \left(\bar{u}(\mathbf{p}'_2 \sigma'_2) \gamma_5 u(\mathbf{p}'_1 \sigma'_1) \right) \left(\bar{u}(\mathbf{p}_2 \sigma_2) \gamma_5 u(\mathbf{p}_1 \sigma_1) \right) \\
 &\times \frac{1}{(p_{1'} - p_1)^2 + m^2 - i\epsilon} \\
 &\quad - [1' \rightleftharpoons 2'], \\
 S_{\mathbf{p}'_1 \mathbf{p}'_2, \mathbf{p}_1 \mathbf{p}_2} &= -(2\pi)^{-6} g^2 (16E_1 E_2 E'_1 E'_2)^{-1/2} \delta^4(p_1 + p_2 - p'_1 - p'_2) \\
 &\times \int d^4 q \operatorname{Tr} \left\{ \gamma_5 \frac{-i\gamma_\mu q^\mu + M}{q^2 + M^2 - i\epsilon} \gamma_5 \frac{-i\gamma_\mu (q + p'_1)^\mu + M}{(q + p'_1)^2 + M^2 - i\epsilon} \right. \\
 &\quad \left. \times \gamma_5 \frac{-i\gamma_\mu (q + p'_1 - p_1)^\mu + M}{(q + p'_1 - p_1)^2 + M^2 - i\epsilon} \gamma_5 \frac{-i\gamma_\mu (q - p'_2)^\mu + M}{(q - p'_2)^2 + M^2 - i\epsilon} \right\} + \dots,
 \end{aligned}$$

where in the last formula the ellipsis indicates a sum over permutations of particles 2, 1', 2'. The factors β in the fermion propagator numerators have been used to replace $u^\dagger$ with $\bar{u}$.

* * *

Another useful topological result expresses a sort of conservation law of lines. For the moment we can think of all internal and external lines as being created at vertices and destroyed in pairs at the centers of internal lines or when external lines leave the diagram. (This has nothing to do with the directions of the arrows carried by these lines.) Equating the